Chapter 1 – Architecture Overview

- MMU purpose and responsibilities
- Two-stage translation model
- Privilege levels
- TLB interaction
- Barriers
- Break-Before-Make (BBM)

Chapter 2 – Addressing & Virtual Address Structure

- VA sizes and canonicality
- TTBR0/TTBR1 split
- Granule size
- VA slicing
- Fault types

Chapter 3 – TTBR, ASID, and Process Isolation

- TTBR0/TTBR1 roles
- ASID tagging and allocation
- VMID interaction
- TTBR updates and barriers

Chapter 4 – TCR, MAIR, and Memory Attribute Control

- TCR fields
- Granule size
- Shareability
- Cacheability
- MAIR encodings
- Stage-1 vs Stage-2 attribute merging

Chapter 5 – Page Table Levels and Descriptor Formats

- Page-table hierarchy
- Descriptor types

- Block vs page mappings
- AF, AP, PXN/UXN, SH, AttrIndx
- Walk behavior

Chapter 6 – Stage-1 Translation (VA → IPA)

- Stage-1 flow
- Descriptor interpretation
- IPA generation
- Permission enforcement
- Fault types

Chapter 7 – Stage-2 Translation (IPA → PA)

- Hypervisor-controlled translation
- S2AP, S2PXN/S2UXN, S2AttrIndx
- PA generation
- VMID isolation

Chapter 8 – TLB Architecture and Behavior

- TLB structure
- ASID + VMID tagging
- Walk cache
- TLBI rules
- BBM interaction

Chapter 9 – Memory Barriers (DSB, DMB, ISB)

- Ordering rules
- Page-table update sequences
- Device ordering
- SMP ordering

Chapter 10 – Break-Before-Make (BBM)

- Why BBM exists
- When BBM is required

- Full BBM sequences
- TLB aliasing prevention

Chapter 11 – Virtualization Pipeline (Stage-1 + Stage-2 Integration)

- Permission merging
- Attribute merging
- Shareability merging
- Virtualization fault behavior

Chapter 12 – Debugging MMU Faults

- ESR/DFSC decoding
- FAR/HPFAR analysis
- Stage-1 vs Stage-2 fault identification
- Debugging workflow

Chapter 13 – Best Practices for MMU Design

- TTBR/TCR/MAIR best practices
- Descriptor layout rules
- ASID/VMID allocation
- SMP and device memory rules

Chapter 14 – Full Translation Pipeline Walkthrough

- End-to-end VA → IPA → PA flow
- TLB insertion
- Final access behavior

Chapter 15 – Memory Types and Cacheability

- Normal vs device memory
- MAIR encodings
- Ordering and speculation rules
- DMA interaction

Chapter 16 – Shareability and Coherency

- Shareability domains
- SMP coherency
- TLB/walk-cache coherence

Chapter 17 – Memory Ordering and Barriers (Advanced)

- Weak memory model
- Ordering domains
- Device ordering
- DMA ordering

Chapter 18 – Cache Maintenance & MMU Interaction

- Cache maintenance instructions
- DMA coherence
- Page-table update rules
- Hypervisor interactions

Chapter 19 – MMU Performance Optimization

- TLB optimization
- Block mapping strategies
- Stage-2 optimization
- SMP scalability

Chapter 20 – Advanced Virtualization Techniques

- Nested virtualization
- Shadow page tables
- Paravirtualization
- Memory overcommit
- Live migration

Chapter 21 – MMU Design Patterns for Real Systems

- Linux/QNX kernel patterns

- Hypervisor patterns
- Embedded/bare-metal patterns
- Safety-critical RTOS patterns
- High-performance server patterns
- Automotive/GPU/TEE patterns

Chapter 22 – Device Mapping & I/O Virtualization

- MMIO rules
- PCIe BAR mapping
- GIC virtualization
- Timer/UART/SPI/I2C/GPIO mapping
- DMA/IOMMU integration
- Device partitioning

Chapter 23 – Interrupts, Exceptions & MMU Interaction

- Exception vector mapping
- Fault delivery paths
- GIC MMU rules
- Interrupt latency and MMU
- Stage-2 interrupt virtualization
- Nested virtualization

Chapter 24 – Boot Process & MMU Bring-Up (EL3 → EL2 → EL1)

- EL3 secure boot MMU setup
- EL2 hypervisor MMU setup
- EL1 kernel MMU setup
- Identity mapping
- Early page-table design
- SMP bring-up

Chapter 25 – Security Architecture & MMU Hardening

- W^X enforcement
- PAN/PXN/UXN/WXN

- TrustZone
- PAC/BTI
- Secure page-table design
- DMA/device isolation

Chapter 26 – Debugging Tools, Tracing & Instrumentation

- Page-table dumpers
- TLB/walk-cache dumpers
- ESR/FAR/HPFAR decoders
- Stage-2 tracers
- Barrier/cache-maintenance tracers
- SMP/hypervisor tracing tools

Chapter 27 – Common MMU Bugs & Failure Modes

- Wrong attributes/shareability/permissions
- Missing AF
- Wrong granule/alignment
- Stale TLB/walk-cache
- Missing barriers
- Stage-2 mapping failures
- Boot identity-mapping failures

Chapter 28 – Full System Case Studies

- Linux
- QNX
- Hypervisors
- Bare-metal
- Automotive ECUs
- Secure enclaves
- Servers
- GPU systems
- Cloud hosts
- Robotics
- Microkernels

- High-security systems
- Multimedia systems
- Multi-tenant isolation

Chapter 29 – Complete MMU Checklist & Reference Tables

- Bring-up checklist
- Descriptor checklist
- MAIR/TCR/TTBR tables
- Permission/shareability/barrier/TLBI rules
- Device mapping rules
- Stage-2 rules
- Security/SMP/debugging/failure-mode tables